

Clinical Trial Complexity Assessment — Quick Reference Card

OPAL / EFACCT Framework · 5 Dimensions · 34 Criteria · Weighted Scoring

Score each item 0–3 | Total = Σ (raw × weight) ÷ 1.245 | Range 0–100

GO = 0–40 · **CONDITIONAL GO** = 41–65 · **NO-GO** = 66–100

#	Criterion	Score 0 — Low / None		Score 1 — Low-Moderate		Score 2 — Moderate-High		Score 3 — High / Critical					
1	Protocol Design & Study Complexity	Weight 1.2×	7 items	Max raw 21	Red flag >15								
1.1	Trial Phase	0	Phase IV / Observational		1	Phase III		2	Phase II		3	Phase I / First-in-Human	
1.2	Study Design Complexity	0	Simple parallel / observational		1	Crossover / factorial		2	Adaptive randomization / dose-finding		3	Complex adaptive with multiple decision points	
1.3	Number of Treatment Arms	0	Single / observational		1	Two arms		2	Three arms		3	Four or more arms	
1.4	Eligibility Criteria Complexity	0	Broad (≤10 criteria)		1	Moderate (11–20 criteria)		2	Restrictive (21–30 criteria)		3	Highly restrictive (>30 or specialized testing required)	
1.5	Biomarker Requirements	0	None required		1	Standard routine laboratory		2	Specialized send-out testing		3	Multiple specialized / novel assays	
1.6	Primary Endpoint Complexity	0	Simple clinical or survival endpoint		1	Patient-reported outcome (PRO) / QoL		2	Composite or imaging endpoint		3	Novel endpoint / central adjudication required	
1.7	Protocol Amendment History	0	Stable — no amendments anticipated		1	1–2 minor amendments		2	3–4 moderate amendments		3	>4 amendments or major design changes	
2	Operational Execution & Site Burden	Weight 1.5×	8 items	Max raw 24	Red flag >17								
2.1	Number of Study Visits	0	≤5 visits		1	6–10 visits		2	11–20 visits		3	>20 visits	
2.2	Visit Duration	0	<1 hour — routine visit		1	1–2 hours — standard procedures		2	2–4 hours — multiple procedures		3	>4 hours — complex / inpatient	
2.3	Assessment Procedures	0	Routine vitals and labs		1	Standard imaging or functional tests		2	Specialized procedures (PK, imaging)		3	Multiple specialized / invasive procedures	
2.4	Data Collection Burden	0	Simple eCRF — minimal fields		1	Standard eCRF — moderate fields		2	Complex eCRF — extensive data points		3	Multiple systems / real-time / PRO support required	
2.5	Monitoring Intensity	0	Remote or annual on-site monitoring		1	Every 6–12 months on-site		2	Every 2–6 months on-site		3	Monthly or 100% SDV (source data verification)	
2.6	Investigational Product Handling	0	Oral drug — simple dispensing		1	Refrigerated storage or reconstitution required		2	IV infusion or complex preparation		3	Specialized storage / dedicated resources required	
2.7	Third-Party Coordination	0	None required		1	Single vendor or CRO		2	Multiple vendors		3	Complex vendor network / decentralized components	
2.8	Safety Reporting Requirements	0	Standard adverse event (AE) reporting		1	Enhanced AE monitoring		2	Real-time SAE / frequent reporting		3	Complex: DSMB / frequent safety labs / specialized reporting	
3	Regulatory & Oversight Requirements	Weight 1.0×	6 items	Max raw 18	Red flag >13								
3.1	IRB / Ethics Review	0	Standard review process		1	Expedited review — minor modifications		2	Full board review — vulnerable populations or sensitive		3	Multi-site or complex consent considerations	
3.2	Regulatory Submissions	0	None required		1	Single submission (e.g., IND amendment)		2	Multiple or specialized submissions		3	Complex regulatory pathway or novel designation	
3.3	Informed Consent Complexity	0	Standard single-signature consent		1	Lengthy consent / additional educational materials		2	Multi-stage consent or legally authorized representative (LAR)		3	Complex — multiple optional sub-studies or ongoing consent	
3.4	Data Privacy Requirements	0	Standard HIPAA compliance		1	Enhanced security protocols		2	International data transfers or specialized requirements		3	Highly sensitive — exceptional security measures required	
3.5	Oversight Committees	0	None required		1	Sponsor safety monitoring only		2	Independent DSMB or adjudication committee		3	Multiple committees / complex governance structure	
3.6	Audit / Inspection Risk	0	Low risk — standard monitoring		1	Moderate risk — possible audit		2	High risk — likely audit		3	Very high risk — anticipated regulatory inspection	
4	Resource & Workforce Capacity	Weight 1.3×	7 items	Max raw 21	Red flag >15								
4.1	Coordinator Time Required	0	<0.25 FTE		1	0.25–0.5 FTE		2	0.5–1.0 FTE		3	>1.0 FTE or multiple coordinators needed	
4.2	Specialized Expertise Needed	0	Standard site capabilities sufficient		1	Brief staff training required		2	Specialized experienced staff needed		3	Highly specialized skills or certification required	
4.3	Space & Equipment Requirements	0	Standard exam room		1	Dedicated research space / standard equipment		2	Specialized equipment or procedure space required		3	Significant infrastructure investment required	
4.4	Pharmacy Resources	0	None or minimal pharmacy involvement		1	Standard pharmacy services		2	Moderate — compounding or specialized storage		3	Intensive / specialized pharmacy capabilities required	
4.5	Laboratory Resources	0	Routine commercial laboratory		1	Standard sample processing and storage		2	Specialized laboratory (e.g., –80°C freezer required)		3	Complex protocols / significant storage investment	
4.6	Budget Adequacy	0	Budget clearly exceeds projected costs		1	Budget adequate with reasonable contingency		2	Marginally adequate — limited contingency		3	Budget insufficient for safe conduct	
4.7	Current Site Workload	0	Significant available capacity		1	Moderate available capacity		2	Limited capacity — near maximum		3	At or over current capacity	
5	Patient Population & Recruitment	Weight 1.0×	6 items	Max raw 18	Red flag >13								
5.1	Target Population Accessibility	0	Large, readily accessible population		1	Moderate — accessible with standard outreach		2	Small — requires active recruitment effort		3	Very limited / rare disease population	
5.2	Enrollment Timeline	0	Relaxed timeline / low enrollment rate		1	Moderate timeline and rate		2	Aggressive timeline / high enrollment rate		3	Very aggressive — likely unachievable	
5.3	Competing Studies	0	No competing studies		1	Minimal competition — distinct populations		2	Moderate competition for same population		3	Direct competition / trial competes with standard of care	
5.4	Patient Burden	0	Minimal burden — convenient participation		1	Moderate burden — manageable		2	Substantial burden — may deter enrollment		3	Excessive burden — significantly impacts consent	
5.5	Retention Challenges	0	Low dropout risk / short study duration		1	Moderate retention challenge		2	Retention challenges anticipated		3	High dropout risk — likely to compromise data	
5.6	Recruitment Strategy Feasibility	0	Proven strategy / high confidence		1	Standard approach / moderate confidence		2	Novel or uncertain strategy		3	Unclear / likely ineffective strategy	

Scoring Formula

Weighted score = raw score × dimension weight

Final score = (Σ weighted) ÷ 1.245 → 0–100 scale

Dimension	Items	Weight	Max Wt.	Red Flag Raw
1. Protocol Design	7	1.2×	25.2	>15
2. Operational Execution	8	1.5×	36.0	>17
3. Regulatory & Oversight	6	1.0×	18.0	>13
4. Resource & Workforce	7	1.3×	27.3	>15
5. Patient & Recruitment	6	1.0×	18.0	>13

Decision Matrix

Overall Score	Dim 5 Raw ≤6	Dim 5 Raw 7–12	Dim 5 Raw 13–18
0–40 (Low)	GO	GO	CONDITIONAL GO
41–65 (Medium)	GO	CONDITIONAL GO	NO-GO
66–100 (High)	CONDITIONAL GO	NO-GO	NO-GO

Automatic NO-GO Override Criteria

- Any single dimension weighted score ≥ 90% of its maximum
- Three or more dimensions with raw score at red-flag threshold
- Item 2.1 (Visits) = 3 AND Item 2.3 (Procedures) = 3 simultaneously
- Item 4.6 (Budget Adequacy) = 3 (budget clearly insufficient)
- Item 4.7 (Workload) = 3 (site at or over capacity)
- Item 5.1 (Population Access) = 3 AND Item 5.2 (Timeline) = 3

Score Thresholds

Band	Score	Decision
LOW	0–40	GO — proceed
MEDIUM	41–65	CONDITIONAL GO
HIGH	66–100	NO-GO — decline

Red flag = any single dimension exceeds its raw threshold listed above. Check overrides before final decision. Use the app at localhost:8080 for automated calculation.